Q 01. what will be the best complexity to check if two strings are anagram or not for given string length of size N.
Example for Anagram: An anagram of a string is another string that contains the same characters, only the order of characters can be different.
str1 = "samsung"
str2 = "gunmass"

Ops: A. O(N^2)

B. O(N*LogN)

C. O(N)

D. O(logN)

Q 02. Which memory sits closest to CPU

Ops: A. L2 Cache

B. TLB

C. L1 Cache

D. RAM

Submit

Sections

Q 03. Consider the 3 processes, P1, P2 and P3 shown in the table.

Process Arrival time Time Units Required

P1 0 5

P2 1 7

P3 3 4

The completion order of the 3 processes under the policies RR2 (round robin scheduling with CPU quantum of 2 time units) are

Ops: A. P1 P2 P3

B. P1 P3 P2

C. P3 P1 P2

D. P3 P2 P1

Q 04. What all can be Cache Replacement Policies

Ops: A. FIFO

B. MRU

C. LRU

D. LIFO

Submit

Q 05. Consider maximum stack size allocated for a program is 5, what is the out of below pseudo code

Q 05. Consider maximum stack size allocated for a program is 5. what is the out of below pseudo code.

```
Initialize Stack(5);
Push(10);
Push(11);
Push(12);
pop();
Push(13);
Push(14);
push(15);
push(16);
pop();
pop();
printf("%d", *top);
```

Ops: A. 12

B. 14

C. stack overflow

D. 10

Submit

Q 06. What will be the output of the following code snippet?

Q 06. What will be the output of the following code snippet?

```
#include <stdio.h>
void calculate() {
    int a = 3;
    int res = a++ + ++a + a++ + ++a;
    printf("%d", res);
}
int main() {
 calculate();
 return 0;
}
```


Ops: A. ○ 18

B. ○ 24

C. ○ 20

D. ○ 16

Q 07. Whats the correct order w.r.t cost of various memory types (Low to High cost)

Submit

Ops: A. ○ CPU registers -> Cache -> RAM -> Hard Disk memory types (Low to High cost)

Ops: A. CPU registers -> Cache -> RAM -> Hard Disk

B. Cache -> CPU Registers -> RAM -> Hard Disk

C. CPU registers -> Cache -> Hard Disk -> RAM

D. Hard Disk -> RAM -> Cache -> CPU Registers

Q 08. what is correct order for 5 stage risc pipeline?

Ops: A. fetch -> decode -> memory access -> execute -> writeback

B. fetch -> memory access -> writeback -> decode execute

C. fetch -> decode -> execute -> memory access -> write back

D. fetch -> memory access -> decode -> execute -> write back

Q 09. Consider we have a function, getLCA(), which returns us the Lowest Common Ancestor between 2 nodes of a tree. Us getLCA() function, how can we calculate the distance between 2 nodes, given that dist() distance from the root, to e calculated?

Ops: A. dist(u) + dist(v) - dist(getLCA(u, v))

Submit

Adithya Nayak
Candidate ID: [illegible]

Q 09. Consider we have a function, getLCA(), which returns us the Lowest Common Ancestor between 2 nodes of a tree. Using this getLCA() function, how can we calculate the distance between 2 nodes, given that dist() distance from the root, to each node is calculated?

Ops: A. ○ dist(u) + dist(v) - dist(getLCA(u, v))

B. ○ dist(u) + dist(v) + 2 * dist(getLCA(u, v))

C. ○ dist(u) + dist(v) + dist(getLCA(u, v))

D. ○ dist(u) + dist(v) - 2 * dist(getLCA(u, v))

Q 10. What will be the output of the following C program?

```
#include <stdio.h>

union values {
    int val1;
    char val2;
} myVal;

int main()
{
    myVal.val1 = 66;
```

tions

Software
33 attempted

Q 10. What will be the output of the following C program?

```c
#include <stdio.h>

union values {
    int val1;
    char val2;
} myVal;

int main()
{
    myVal.val1 = 66;

    printf("val1 = %p", &myVal.val1);
    printf("\nval2 = %p", &myVal.val2);

    return 0;
}
```


Ops: A. ○ val1 = 0x55ac76dd2014
val2 = 0x55ac76dd2014

Ops: A. val1 = 0x55ac76dd2014
val2 = 0x55ac76dd2014

B. val1 = 0x54ac88dd2012
val2 = 0x55ac76dd2014

C. Compile Error

D. Run time error

Q 11. The prefix form of A-B/ (C * D ^ E) is?

Ops: A. -ABCD*^DE

B. -A/B*C^DE

C. -/*^ACBDE

D. -A/BC*^DE

Q 12. p is the probability that event X occurs, how do we find odds ratio for this event

Ops: A. p/(1-p)

Submit

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Ops: A. ○ -ABCD*^DE

B. ○ -A/B*C^DE

C. ○ -/*^ACBDE

D. ○ -A/BC*^DE

Q 12. p is the probability that event X occurs, how do we find odds ratio for this event

Ops: A. ○ p/(1-p)

B. ○ 1/(1-p)

C. ○ (1-p)/p

D. ○ 1-p

Q 13. Which one of the following is FALSE?

Ops: A. ○ Context switching between user level threads is faster than context switching between kernel level

B. ○ Kernel level threads cannot share the code segment.

Submit

Q 13. Which one of the following is FALSE?

Ops: A. Context switching between user level threads is faster than context switching between kernel level threads.

B. Kernel level threads cannot share the code segment.

C. When a user level thread is blocked, all other threads of its process are blocked.

D. User level threads are not scheduled by the kernel.

Q 14. What is the 16-bit compiler allowable range for integer constants?

Ops: A. -32768 to 32767

B. -3.4e38 to 3.4e38

C. -32668 to 32667

D. -32767 to 32768

Q 15. What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be main[illegible] sorted order ?

Ops: A. $\Theta(1)$

Q 15. What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order ?

Ops: A. ○ $\Theta(1)$

B. ○ $\Theta(n \log n)$

C. ○ $\Theta(n)$

D. ○ $\Theta(n2)$

Q 16. A thread is usually defined as a "light weight process" because an operating system (OS) maintains smaller data structures for a thread than for a process. In relation to this, which of the following is TRUE?

Ops: A. ○ On per-thread basis, the OS maintains only scheduling and accounting.

B. ○ On per-thread basis, the OS does not maintain virtual memory state.

C. ○ On per-thread basis, the OS maintains only CPU register state.

D. ○ The OS does not maintain a separate stack for each thread.

Q 17. Which of the below protocol used to perform Ping operation

Q 17. Which of the below protocol used to perform Ping operation

Ops: A. TCP

B. HTTP

C. ICMP

D. DNS

Q 18. A spectrum of 30 MHz is allocated to a cellular system which uses two 25 KHz simplex channels to provide full duplex voice channels. What is the number of channels available per cell for 4 cell reuse factor?

Ops: A. 100 channels

B. 200 channels

C. 300 channels

D. 150 channels

Q 19. What will be the output of the following C code?

```
#include <stdio.h>

int main()
{
```

Q 19. What will be the output of the following C code?

```c
#include <stdio.h>

int main()
{
    char str1[] = "Hello";
    char str2[10];

    str2 = str1;

    printf("%s,%s", str1, str2);

    return 0;
}
```


Ops: A. Hello,

B. Hello,HelloHello

C. Hello,Hello

D. Error

omit

C. Hello,Hello

D. Error

Q 20. ______________ is a condition in which the memory is dynamically reserved but isn't accessible to any program.

Ops: A. Pointer Leak

B. Dangling Pointer

C. Memory Leak

D. Frozen Memory

Q 21. Worst case time complexity to access an element in a BST can be?

Ops: A. O(n)

B. O(logn)

C. O(1)

D. O(n * logn)

Submit

D. O(n * logn)

Q 22. What will be the output of the following C program?

```
#include <stdio.h>

int main()
{
    int x[5] = { 10, 20, 30 };
    printf("%d", x[-1]);
    return 0;
}
```

Ops: A. 10

B. 0

C. Garbage value

D. Error

Q 23. Which algorithm is not part of Supervised learning model?

Ops: A. k-means

D. Error

Q 23. Which algorithm is not part of Supervised learning model?

Ops: A. k-means

B. KNN

C. Linear Regression

D. Logistic regression

Q 24. How much percentage of memory will be saved if we use bit-fields for the given C structure as compared to when we don't use bit-fields for this very structure? (Assume the size of int to be 4).

```
struct temp
{
    int m : 1;
    int n : 2;
    int o : 4;
    int p : 4;
}s;
```


Submit

```
    int p : 4;
}s;
```

Ops: A. 75%

B. 25%

C. 50%

D. 33.30%

Q 25. What is Overfitting in machine learning

Ops: A. Model to automate the manual tasks is called overfitting.

B. It enables computers to learn from data

C. Model performs well on training data and performs well on new data.

D. Model performs well on training data and performs poorly on new data.

Q 26. The following C program is executed on a Unix/Linux system:

```

#include < unistd.h >
int main ()
{
```

Submit

D. ○ 13.30%

Q 25. What is Overfitting in machine learning

Ops: A. ○ Model to automate the manual tasks is called overfitting.

B. ○ It enables computers to learn from data

C. ○ Model performs well on training data and performs well on new data.

D. ○ Model performs well on training data and performs poorly on new data.

Q 26. The following C program is executed on a Unix/Linux system:

```c

#include < unistd.h >
int main ()
{
    int i ;
    for (i=0; i<10; i++)
     if (i%2 == 0) fork ( ) ;
    return 0 ;
}
```

Submit

The total number of child processes created are ____.

C. ○ Model performs well on training data and performs well on new data.

D. ○ Model performs well on training data and performs poorly on new data.

Q 26. The following C program is executed on a Unix/Linux system:

```c

#include < unistd.h >
int main ()
{
    int i ;
    for (i=0; i<10; i++)
     if (i%2 == 0) fork ( ) ;
    return 0 ;
}
```


The total number of child processes created are ______.

Ops: A. ○ 25

B. ○ 32

C. ○ 31

D. ○ 5

D. 5

Q 27. What is the output of below code.

```
int *funcA(void)
{
        int a =20;
        return &a;
}

int main(void)
{
        int *b;
        b= funcA();
        printf("%d", *b);
}
```


Ops: A. Segmentation Fault

B. 20

C. 0x20

D. Compile Error

Submit

D. O Compile Error

Q 28. Which of the following statements are true?

I. Shortest remaining time first scheduling may cause starvation

II. Preemptive scheduling may cause starvation

III. Round robin is better than FCFS in terms of response time

Ops: A. O II and III only

B. O I only

C. O I, II and III

D. O I and II only

Q 29. The maximum number of processes that can be in Ready state for a computer system with n CPUs is

Ops: A. O 2^n

B. O n^2

C. O Independent of n

D. O n

Submit

Q 30. In a connected graph of n nodes and n edges, how many cycles will be present?

Ops: A. Atmost 2

B. Depends on graph

C. nLogn

D. Exactly 1

Q 31. Assume the child thread ran first

```
#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <unistd.h>

int main()
{
int x = 1, y = 1;
pid_t p = fork();
if(p<0){
perror("fork fail");
exit(1);
```

Submit

#include <sys/types.h>
#include <unistd.h>

```
int main()
{
int x = 1, y = 1;
pid_t p = fork();
if(p<0){
perror("fork fail");
exit(1);
}
else if (p == 0)
printf("Child has x = %d y = %d\n", ++x, --y);
else
printf("Parent has x = %d y = %d\n", --x, ++y);

return 0;
}
```


Ops: A. ○ Child has x = 2 y = 0
Parent has x = 0 y = 2

B. ○ Child has x = 2 y = 0
Parent has x = 1 y = 1

C. ○ Since X & Y are shared variables, data synchonization mechanism should be used

Submit

C. Since X & Y are shared variables, data synchronization mechanism should be used

D. either A or B

Q 32. what does below code does and what is the complexity of below code.

```
#include <iostream>
#include <string>
#include <queue>
using namespace std;

void generate(int n)
{
queue<string> q;
q.push("1");

int i = 1;
while (i++ <= n)
{
q.push(q.front() + "0");
q.push(q.front() + "1");

cout << q.front() << ' ';
q.pop();
}
}
```


Submit

```
q.push(q.front() + "0");
q.push(q.front() + "1");

cout << q.front() << " ";
q.pop();
}
}

int main()
{
int n = 4;
generate(n);

return 0;
}
```

Ops: A. ○ 1 10 11 100 101
Complexity O(logN)

B. ○ 1 2 3 4
Complexity O(N^2)

C. ○ 1 2 3 4
Complexity O(N)

D. ○ 1 10 11 100
Complexity O(N)

Submit

Complexity O(logN)

B. 1 2 3 4
Complexity O(N^2)

C. 1 2 3 4
Complexity O(N)

D. 1 10 11 100
Complexity O(N)

Q 33. If the memory access time is denoted by 'ma' and 'p' is the probability of a page fault (0 <= p <= 1). Then the effective acce
for a demand paged memory is ________

Ops: A. None of these

B. (1-p) x ma + p x page fault time

C. p x ma + (1-p) x page fault time

D. ma + page fault time

Submit

SECTION 00/01
Previous Section

SECTION 01/01
Next Section

Problem Statement

5 marks

A robot wants to pick strawberries from a strawberry bush.
You are given a function,

```
int MaximumStrawberries(int* arr, int n, int num);
```

The function accepts an integer array **arr** of length **n** and an integer **num** as its argument where **num** is the number of strawberries a robot will pick as maximum. Each array elemen **arr** represents number of strawberries present in each bush and **n** is the number of bushes. A robot cannot pick strawberries from two consecutive bushes and it has to pick strawberries in such a way that it may not exceed the limit proposed i.e., **num**. Implement the function to find and return the maximum number of strawberries a robot can pick.

Assumption: **n** > 0 and **num** > 0

Note:

- Return 0 if it is not possible to pick strawberries within maximum limit **num**.
- The robot can pick either all the strawberries from each bush or zero of them, i.e., the robot cannot pick partial amount of strawberries.

Constraints:

- $0 <$ number of bushes ≤ 10000
- $0 <$ maximum number of strawberries a robot can pick ≤ 1000
- $0 <$ number of strawberries present in each bush ≤ 1000

Example:

Input:

100
50 10 20 30 40

Output:

90

Explanation:

A robot picks strawberries from the 0^{th} and the 4^{th} bush, total of $(50 + 40 = 90)$ strawberries. It could pick the 2^{nd} bush, but it would exceed the limit (100). Thus, outp

The custom input format for the above case:

100

5

0 10 20 30 40

he first line represents 'num', the second line represents 'n', the third line represents 'arr')

nple input

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The custom input format for the above case:

100
5
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Sample input

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50 10 20 30 40
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The custom input format for the above case:
100
5
50 10 20 30 40
(The first line represents 'num', the second line represents 'n', the third line represents 'arr')

Sample input

```
10
15 20
```

Sample Output

```
0
The custom input format for the above case:
10
2
15 20
(The first line represents 'num', the second line represents 'n', the third line represents 'arr')
```


Instructions :

- This is a template based question, DO NOT write the "main" function.
- Your code is judged by an automated system, do not write any additional welcome/greeting messages.
- "Save and Test" only checks for basic test cases, more rigorous cases will be used to judge your code while scoring.
- Additional score will be given for writing optimized code both in terms of memory and execution time.

Now let's start coding :

Language: C (Gcc 11.3)

01.

15 20

(The first line represents 'num', the second line represents 'n', the third line represents 'arr')

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Now let's start coding :

Language: C++ (Gcc 11.3

Read-only code below . . .

```
int MaximumStrawberries(int* arr, int n, int num);

int main() {
    //Input read from STDIN
    int result = MaximumStrawberries(arr, n, num);
    //Value in result is printed to STDOUT
    return 0;
}

```

Write your code below . . .

```
int MaximumStrawberries(int* arr, int n, int num) {
    /* Write your code here. */
}



```